

- **5-Input Majority Voting
Logic Design**

presented by

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Introduction & Problem

5 voters $\rightarrow$ Yes (1) / No (0)

Output = 1 if ≥ 3 Yes

Inputs A,B,C,D,E

Output: F

System States

$$\text{Total states} = 2^5 = 32$$

$$16 \rightarrow F = 1/16 \rightarrow F = 0$$

Output = 1 when:
3, 4, or 5 Yes votes

Karnaugh Map

K-Map for A = 0

BC	00	0	0	0	0
	01	0	0	1	0
	11	0	1	1	1
	10	0	0	1	0
		00	01	11	10
		AB			

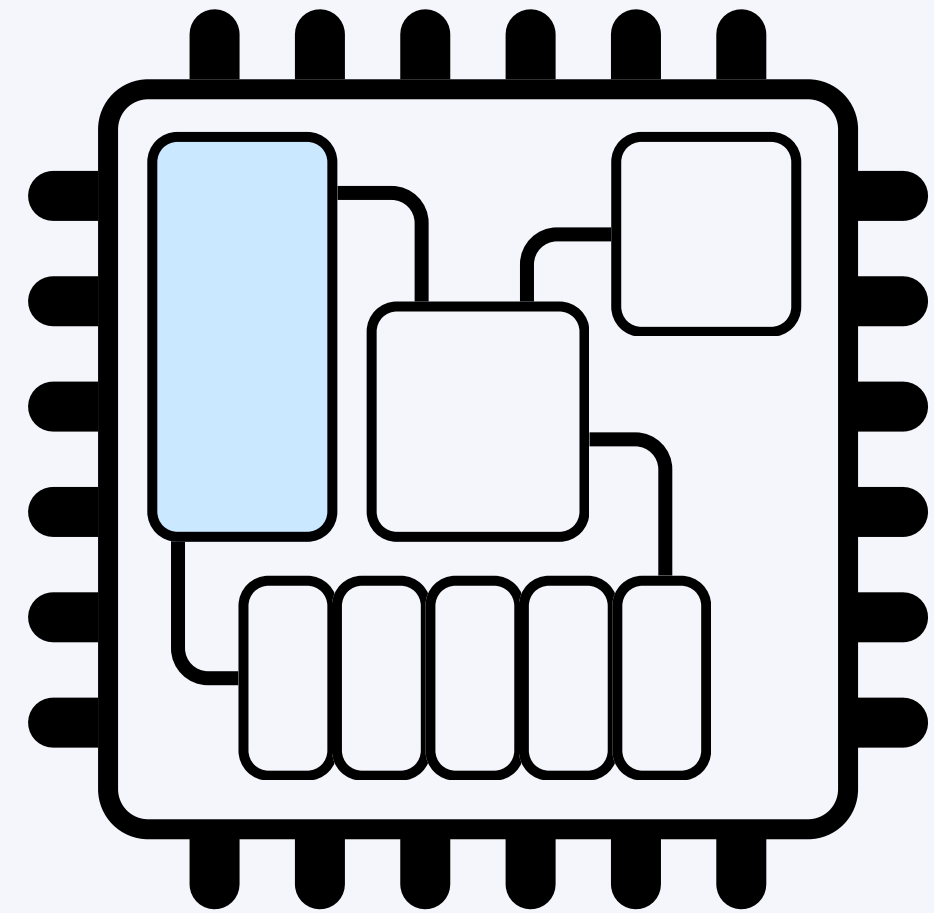
K-Map for A = 1

BC	00	0	0	1	0
	01	0	1	1	1
	11	1	1	1	1
	10	0	1	1	1
		00	01	11	10
		AB			

Boolean Logic

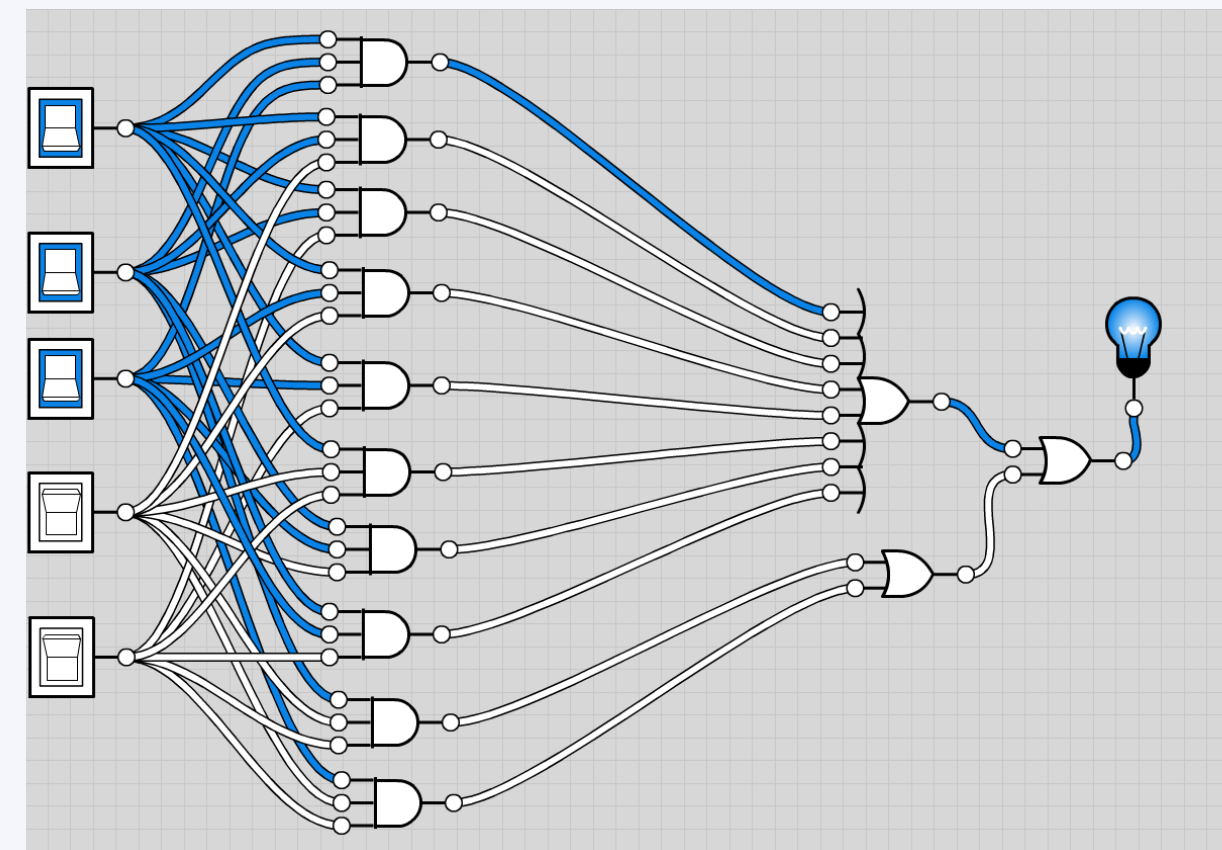
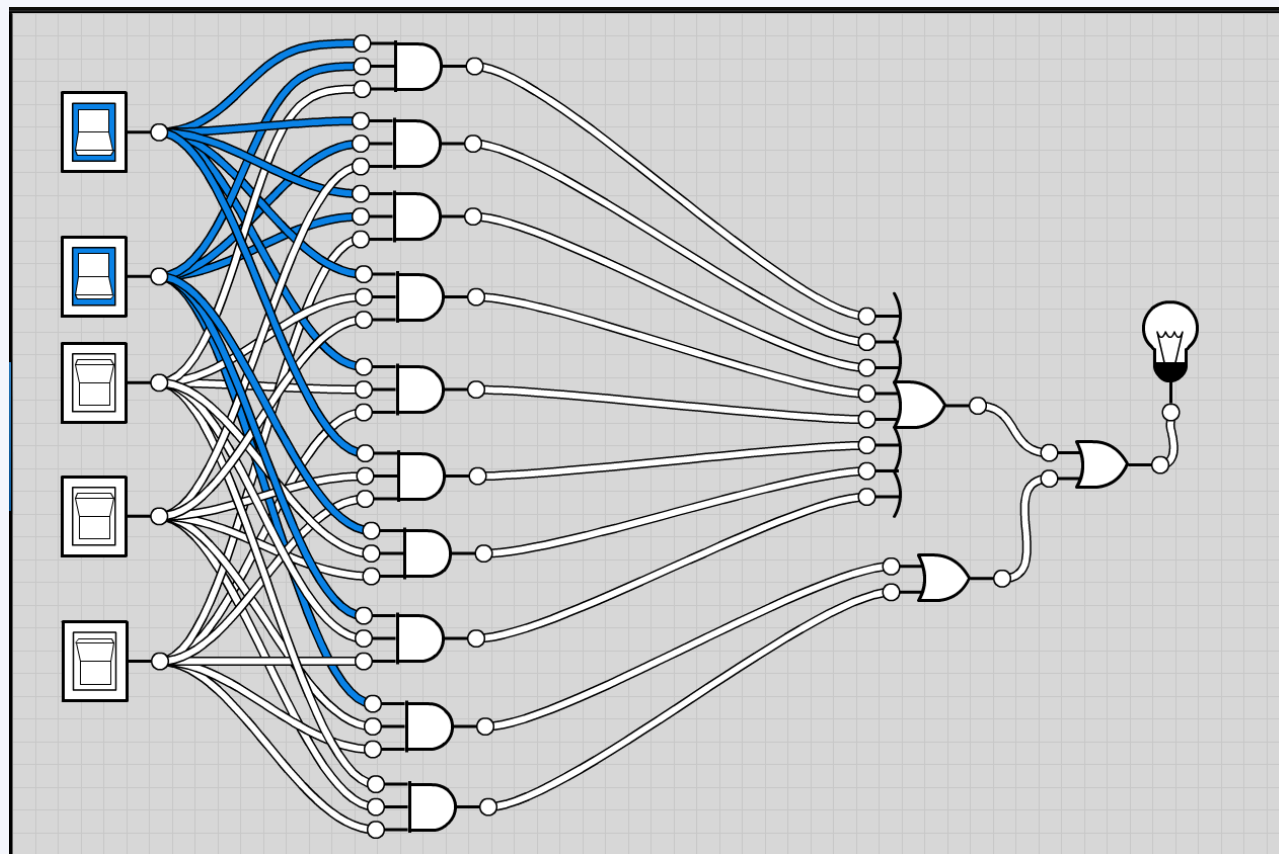
SOP concept: any 3 inputs = 1

$$F = ABC + ABD + ABE + ACD + ACE \\ + ADE + BCD + BCE + BDE + CDE$$



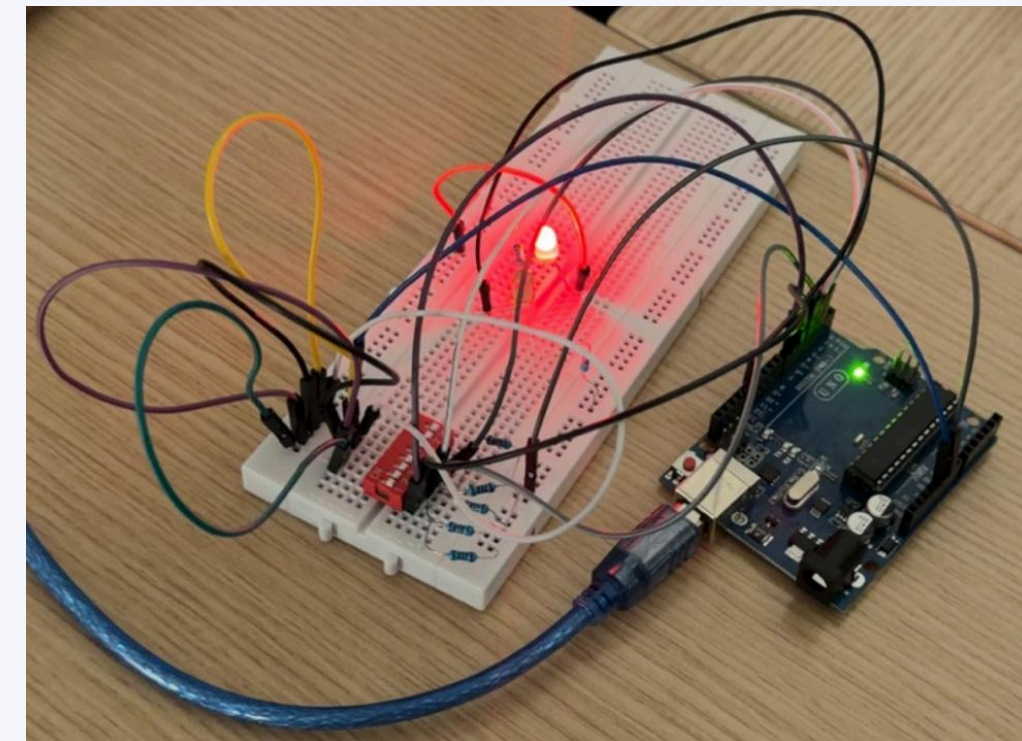
Logic Circuits

- AND gates
- OR gates



Arduino Components

- **Arduino Uno Board**
Used as the main controller of the system.
- **Push Buttons (5 Buttons)**
Used as input switches to represent logic inputs.
- **LEDs (2 LEDs: Red & Green)**
Used to indicate the output result.
- **Resistors (7 Resistors – 10k Ω & 220 Ω)**
10k Ω resistors for pull-down with buttons
220 Ω resistors to protect LEDs
- **Breadboard**
Used for easy and safe circuit connections.
- **Jumper Wires (Male-Male)**
Used to connect components together.
- **USB Cable**
Used to upload the code to the Arduino board.



Logic code

```
int A = 2;
int B = 3;
int C = 4;
int D = 5;
int E = 7;

int LED = 13;

void setup() {
  pinMode(A, INPUT);
  pinMode(B, INPUT);
  pinMode(C, INPUT);
  pinMode(D, INPUT);
  pinMode(E, INPUT);

  pinMode(LED, OUTPUT);
}

void loop() {
  int count = 0;

  if (digitalRead(A) == HIGH) count++;
  if (digitalRead(B) == HIGH) count++;
  if (digitalRead(C) == HIGH) count++;
  if (digitalRead(D) == HIGH) count++;
  if (digitalRead(E) == HIGH) count++;

  if (count >= 3) {
    digitalWrite(LED, HIGH);
  } else {
    digitalWrite(LED, LOW);
  }
}
```